

7(a) — Abstract

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Thesis Title: Advanced Deep Learning Architectures and Adaptive Frameworks for Medical Image Restoration

Medical images acquired using X-ray and computed tomography (CT) are often degraded by acquisition noise and patient motion, reducing diagnostic accuracy. Deep learning methods have shown promise for image restoration, but existing approaches have limitations. Standard denoisers often smooth out fine anatomical details. Most activation functions apply the same operation everywhere, without considering local context.

This thesis addresses these limitations through a three-phase investigation. Phase I presents **X-GAN**, a physics-informed generative adversarial network for chest X-ray denoising, achieving a PSNR of 36.48 dB and an Edge Preservation Index of 0.721 with only 2.17 million parameters. Phase II introduces **SAGA**, a spatially adaptive gated activation function improving PSNR by 6.3 dB over ReLU on average across four architectures. Phase III will develop **Viveka**, a test-time refinement framework, and **ADL₁**, a Sobolev-regularised loss function with an attention-guided U-Net that guarantees gradient fidelity.

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